

HW1_25

2026-03-09

探討「家庭所得」(INCOME) 與「在外用餐支出」(FOODAWAY) 之間的關係。

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':  
##  
##   filter, lag
```

```
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
setwd("/Users/chenzhiyi/Documents/碩班/碩一/財務計量經濟/HW1/")  
cex <- read.csv("cex5_small.csv")
```

```
#Check structure  
str(cex)
```

```
## 'data.frame':   1200 obs. of  10 variables:  
## $ advanced: int  0 0 0 0 0 1 0 0 0 0 ...  
## $ alcbev  : num  11.1 0 10 11.1 0 ...  
## $ appar   : num  0 6.22 0 29.44 22.44 ...  
## $ college : int  0 0 1 1 1 0 0 0 0 0 ...  
## $ entert  : num  43.4 17.6 5 20 56 ...  
## $ food    : num  9.63 94.22 38.52 173.33 173.33 ...  
## $ foodaway: num  47.6 77 39.8 0 108.3 ...  
## $ health  : num  58.22 3.33 33.33 130.59 135.11 ...  
## $ income  : num  123.7 59.7 81 133.2 81 ...  
## $ smsa    : int  1 1 1 1 0 1 1 1 1 1 ...
```

```
summary(cex)
```

```
##      advanced      alcbev      appar      college
## Min.      :0.0000  Min.      : 0.000  Min.      : 0.00  Min.      :0.0000
## 1st Qu.:0.0000  1st Qu.: 0.000  1st Qu.: 0.00  1st Qu.:0.0000
## Median :0.0000  Median : 0.000  Median : 11.11  Median :0.0000
## Mean    :0.2142  Mean    : 7.158  Mean    : 27.53  Mean    :0.3075
## 3rd Qu.:0.0000  3rd Qu.: 7.027  3rd Qu.: 35.56  3rd Qu.:1.0000
## Max.    :1.0000  Max.    :333.330  Max.    :575.33  Max.    :1.0000
##      entert      food      foodaway      health
## Min.      : 0.00  Min.      : 9.63  Min.      : 0.00  Min.      : 0.00
## 1st Qu.: 12.08  1st Qu.: 57.78  1st Qu.: 12.04  1st Qu.: 8.33
## Median : 27.39  Median : 99.80  Median : 32.55  Median : 48.95
## Mean    : 49.82  Mean    :114.44  Mean    : 49.27  Mean    : 84.15
## 3rd Qu.: 63.25  3rd Qu.:145.00  3rd Qu.: 67.50  3rd Qu.:103.58
## Max.    :814.33  Max.    :476.67  Max.    :1179.00  Max.    :1402.89
##      income      smsa
## Min.      : 10.00  Min.      :0.0000
## 1st Qu.: 40.00  1st Qu.:1.0000
## Median : 65.29  Median :1.0000
## Mean    : 72.14  Mean    :0.8875
## 3rd Qu.: 96.79  3rd Qu.:1.0000
## Max.    :200.00  Max.    :1.0000
```

對象：三人家庭（夫、妻、外加一人），月收入在 \$1,000 至 \$20,000 之間。

FOODAWAY：過去一季每人每月的平均在外用餐支出（美元）。

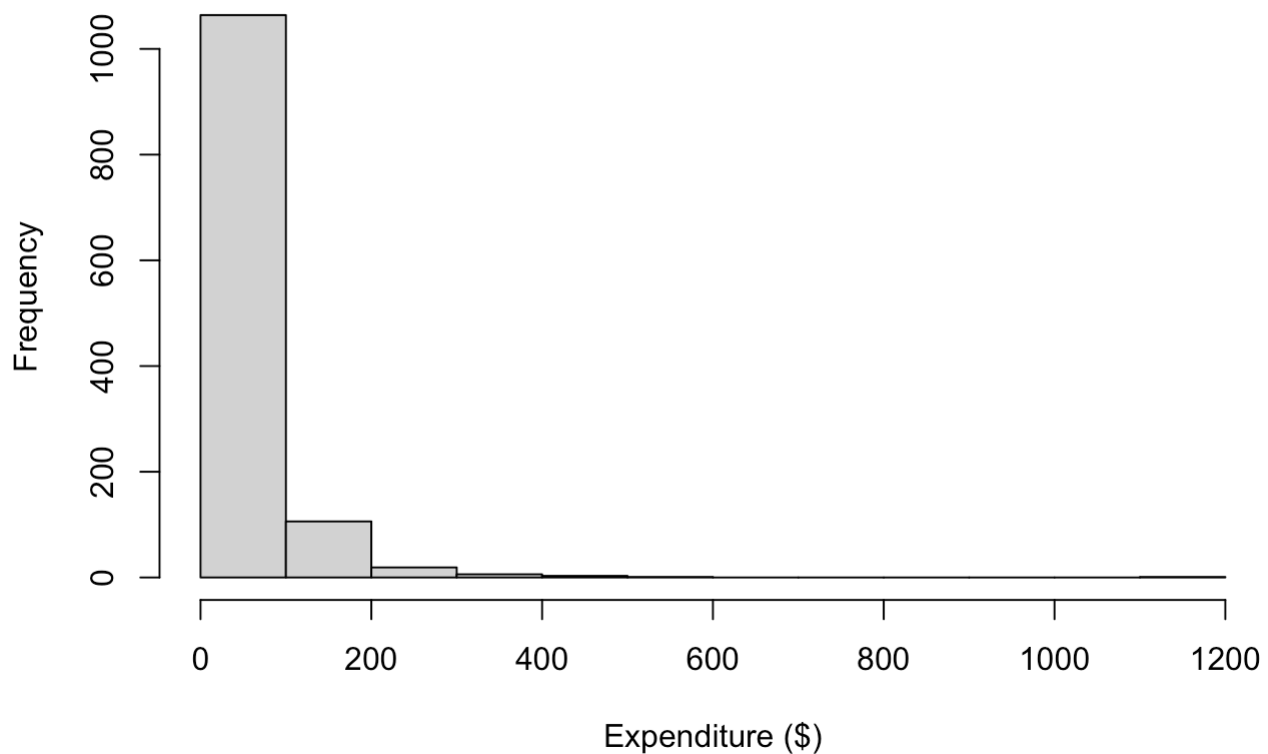
INCOME：過去一年的家庭月收入，單位為 100 美元（例如：數值 50 代表月入 \$5,000）。

(a)

直方圖與敘述統計

```
hist(cex$foodaway, main="Histogram of FOODAWAY", xlab="Expenditure ($)")
```

Histogram of FOODAWAY



```
summary_food <- summary(cex$foodaway)
quantile_food <- quantile(cex$foodaway, probs = c(0.25, 0.75), na.rm = TRUE)

print(summary_food)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.00   12.04   32.55   49.27   67.50  1179.00
```

```
print(quantile_food)
```

```
##      25%      75%
## 12.0400 67.5025
```

```
cat("Valid N =",sum(!is.na(cex$foodaway)))
```

```
## Valid N = 1200
```

分佈右偏，大多分布在100以下。

(b)

不同學歷背景的支出比較

```
group_stats <- cex %>%
  group_by(advanced, college) %>% # 根據實際資料欄位調整
  summarise(
    Mean = mean(foodaway, na.rm = TRUE),
    Median = median(foodaway, na.rm = TRUE)
  )
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by advanced and college.
## i Output is grouped by advanced.
## i Use `summarise(groups = "drop_last")` to silence this message.
## i Use `summarise(by = c(advanced, college))` for per-operation grouping
## (`?dplyr::dplyr_by`) instead.
```

```
print(group_stats)
```

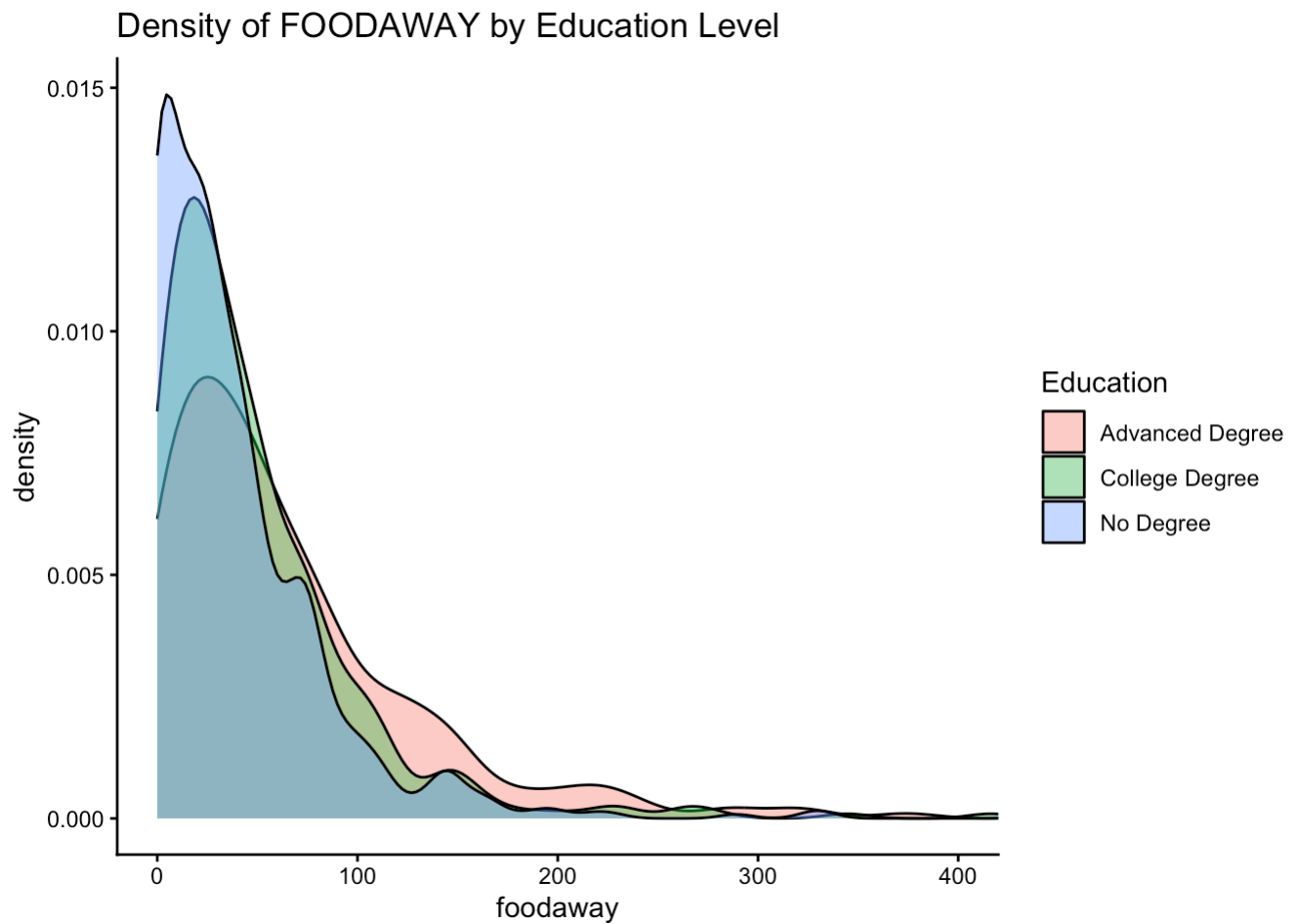
```
## # A tibble: 3 × 4
## # Groups:   advanced [2]
##   advanced college Mean Median
##   <int>    <int> <dbl> <dbl>
## 1      0      0  39.0  26.0
## 2      0      1  48.6  36.1
## 3      1      0  73.2  48.2
```

```
library(ggplot2)
```

```
# 1. 建立學歷標籤
```

```
cex_plot <- cex %>%
  mutate(Education = case_when(
    advanced == 1 ~ "Advanced Degree",
    college == 1 ~ "College Degree",
    TRUE ~ "No Degree"
  ))
```

```
ggplot(cex_plot, aes(x = foodaway, fill = Education)) +
  geom_density(alpha = 0.4) +
  coord_cartesian(xlim = c(0, 400)) +
  labs(title = "Density of FOODAWAY by Education Level") +
  theme_classic()
```



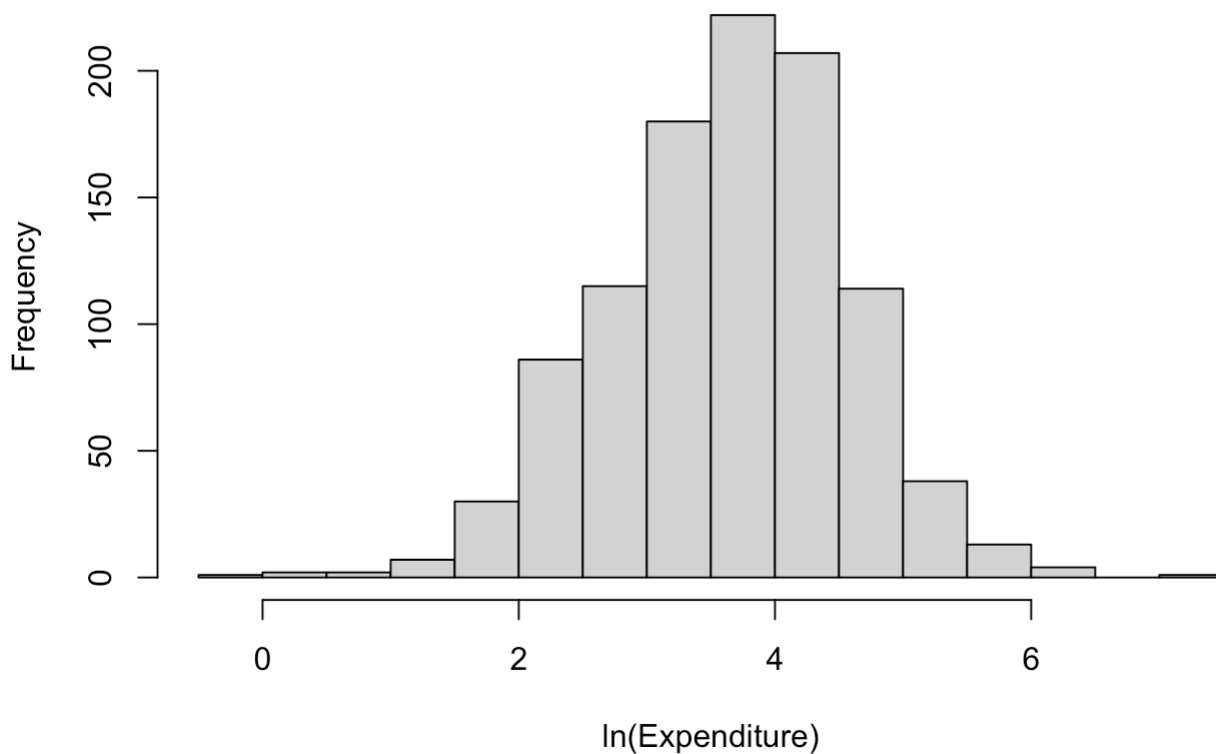
隨著學歷越高，食物支出的中位數和平均也越高。

(c)

對數轉換與樣本數差異

```
cex_new <- subset(cex, foodaway > 0)
hist(log(cex_new$foodaway), main="Histogram of ln(FOODAWAY)", xlab="ln(Expenditure)")
```

Histogram of ln(FOODAWAY)



```
# 排除 -Inf
summary(log(cex_new$foodaway))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -0.3011  3.0759  3.6865  3.6508  4.2797  7.0724
```

```
# 計算真正參與回歸的樣本數
cat(" Valid N = ", length(cex_new$foodaway))
```

```
## Valid N = 1022
```

取對數後偏向常態分佈。排除了NaN值 -inf (log(0))後，有效樣本數只剩1022。

(d)

線性回歸。

```
model_d <- lm(log(foodaway) ~ income, data = cex_new)
summary(model_d)
```

```
##
## Call:
## lm(formula = log(foodaway) ~ income, data = cex_new)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.6547 -0.5777  0.0530  0.5937  2.7000
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.1293004   0.0565503   55.34  <2e-16 ***
## income       0.0069017   0.0006546   10.54  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8761 on 1020 degrees of freedom
## Multiple R-squared:  0.09826,    Adjusted R-squared:  0.09738
## F-statistic: 111.1 on 1 and 1020 DF,  p-value: < 2.2e-16
```

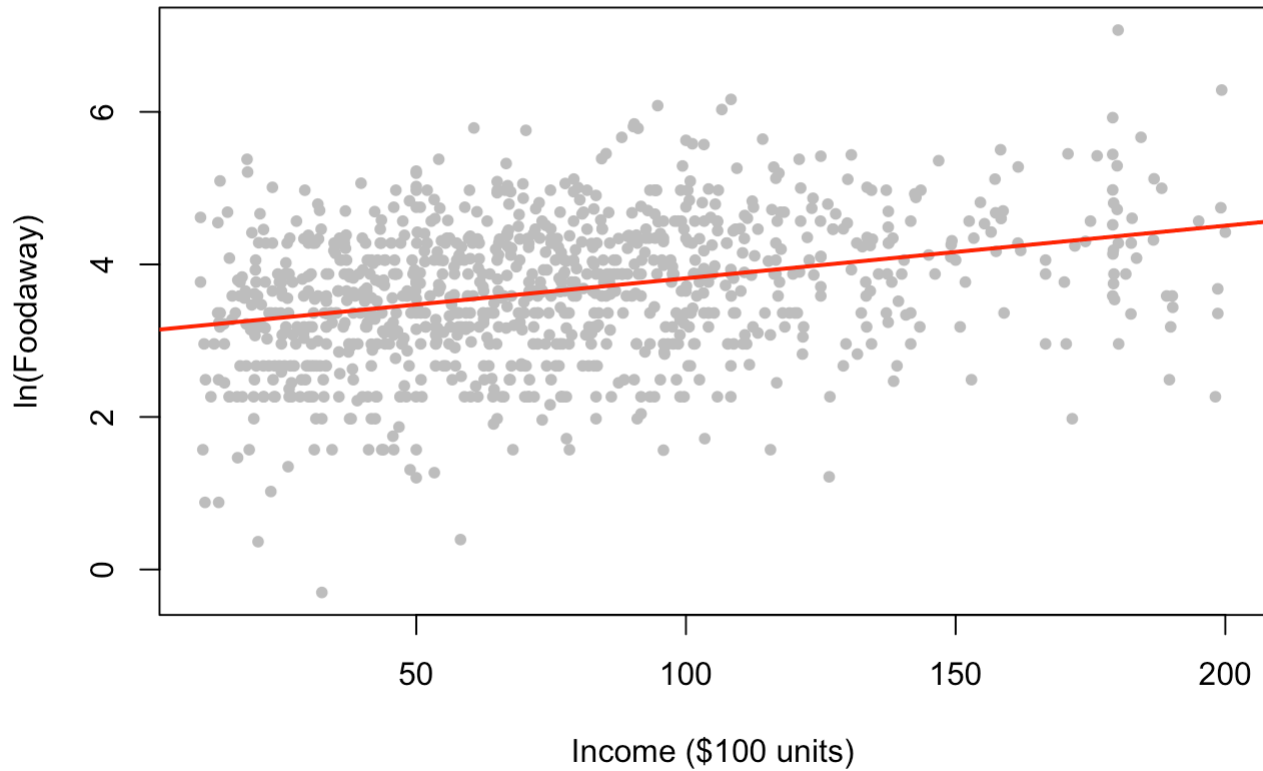
$$\ln(\widehat{FOODAWAY}) = 3.1293 + 0.0069 \times INCOME$$

1. $\beta_2 = 0.0069$ ，在其他條件不變的情況下，家庭月收入每增加 1 單位（即 100 美元），其每人每月的在外用餐支出預計增加約 0.69%。
2. $P - value < 2e - 16 < 0.05$ ，代表income 對 $\ln(\text{foodaway})$ 有極其顯著的正向影響。

(e)

Scatter plot and fitted line

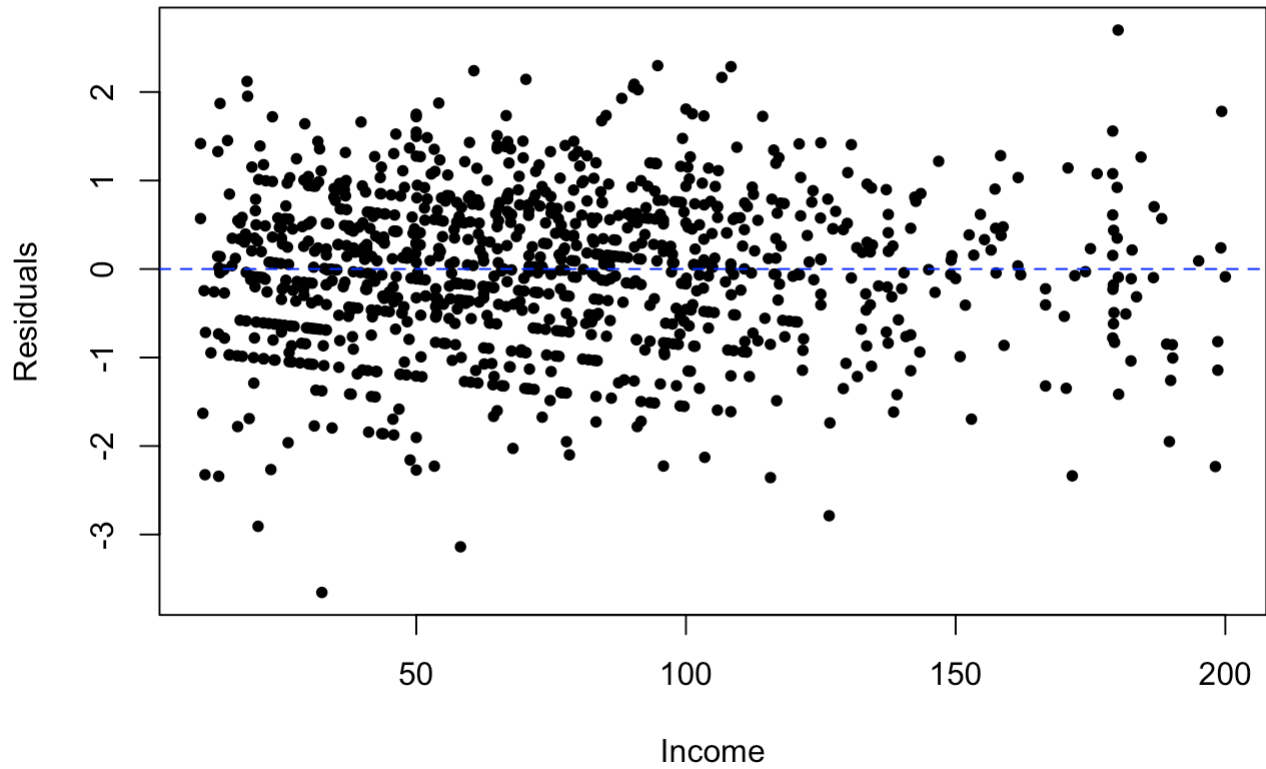
```
plot(cex_new$income, log(cex_new$foodaway), pch=20, col="gray",
     xlab="Income ($100 units)", ylab="ln(Foodaway)")
abline(model_d, col="red", lwd=2)
```



(f) 殘差分析

```
cex_new$residuals <- resid(model_d)
plot(cex_new$income, cex_new$residuals, pch=20,
     main="Residuals vs Income", xlab="Income", ylab="Residuals")
abline(h=0, lty=2,col='blue')
```


Residuals vs Income



殘差在水平線上下的分布平均。